

Sohith Bandari

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EDUCATION

New York University

September 2024 - May 2026

Master of Science in Computer Science (GPA: 3.8)

New York, United States

Relevant coursework: Machine Learning, Artificial Intelligence, Deep Learning, MLOps, Software Engineering

Indian Institute of Technology, Dhanbad

December 2020 - May 2024

Bachelor of Technology in Electronics and Communication Engineering

Jharkhand, India

Relevant coursework: Probability & Statistics, Data Mining, Digital Image Processing, Pattern Recognition

TECHNICAL SKILLS

Languages: C/C++, Python, SQL, MATLAB, Rust

Frameworks: PyTorch, TensorFlow, JAX, LangChain, Streamlit

Libraries: NumPy, Pandas, Matplotlib, sciPy, scikit-learn, Keras, HuggingFace

Tools: Git, GitHub, Linux, AWS, Docker, Kubernetes, FastAPI, Excel, PostgreSQL

EXPERIENCE

Reliance Jio Infocomm Limited | Software Engineering Intern (June 2023 - July 2023)

- Implemented an end-to-end Computer Vision model to detect empty spaces in data center racks using YOLOv8m. The dataset was derived from images on the Internet & labelled manually, then preprocessed & augmented for better results.
- Testing the model yielded a score of mAP50: 0.517 and mAP50-95: 0.312.

- Tested multiple models for the team on Gradio which returned satisfactory results. This model will be assessed on real-world data and deployed on a production scale.

PROJECTS

HerWay: Women Safety App | PyTorch, Huggingface, FastAPI

- Provided a comprehensive solution to the problem of harassment against women in public space.
- Leveraged sentence analysis on data from safecity.in to assess location safety. Employed geocoding for coordinates retrieval and calculated risk based on nearby crime severity. Utilised graph algorithms to recommend the safest routes for women travelers.
- 2nd Runner-up in HackFest 2023. The project was discussed during a session in the Rajya Sabha, the upper house of India's Parliament.

Git Issue Bot | Python, LangChain, Streamlit, Docker, PostgreSQL

- Engineered an intelligent chatbot leveraging OpenAI's API to dynamically search GitHub repositories and issues.
- Architected a robust conversational system using LangChain framework, implementing custom tools and chains for relevant context-aware responses and improved search accuracy.
- Designed and implemented a PostgreSQL database system for efficient management of user interactions, chat histories, and personalized repository and issue bookmarking.
- Developed a containerized full-stack application using Streamlit for the frontend interface, ensuring seamless deployment and scalability through Docker.

RAG System for Code Generation | Python, Huggingface, ClearML, MongoDB, Qdrant

- Implemented a RAG system to assist software developers, enabling domain-specific query responses for coding tasks.
- Created an ETL pipeline from scratch in Python to ingest and process data from C++ and Python documentations, storing raw data in MongoDB and vectorized representations in Qdrant.
- Fine-tuned LLaMa 3.2 3b by implementing LoRA (Low-Rank Adaptation) on a custom preference dataset generated using chatGPT API.
- Deployed a Gradio-based application with Huggingface Hub and Ollama integration for code generation instructions. Utilized Open WebUI for an enhanced user interface and improved interaction with the LLM model.